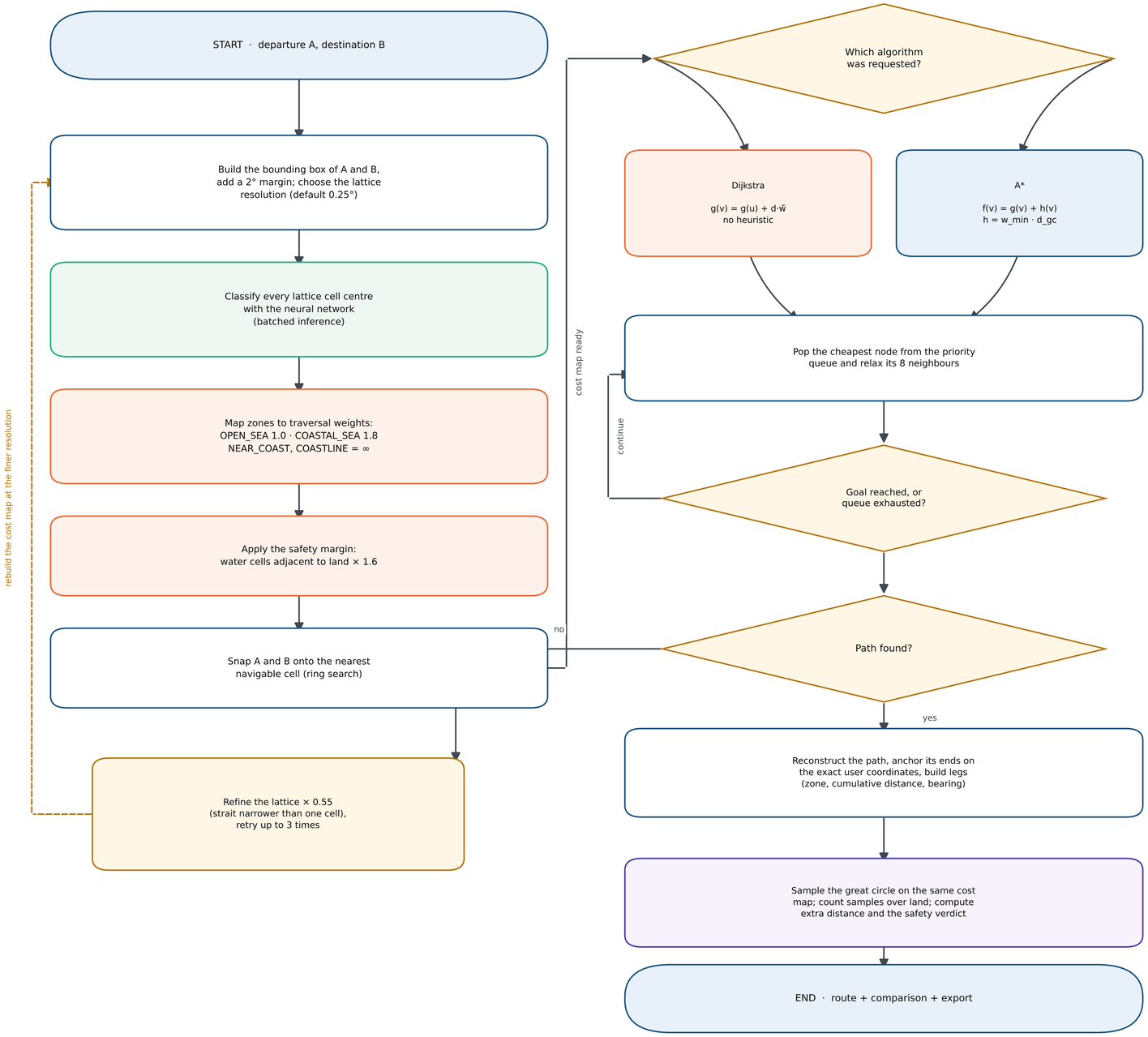
Algorithm of route construction

Flowchart of cost-map generation and A* / Dijkstra search

Sheet 5

Format A1



Cost model

Traversal cost of the edge (u → v):

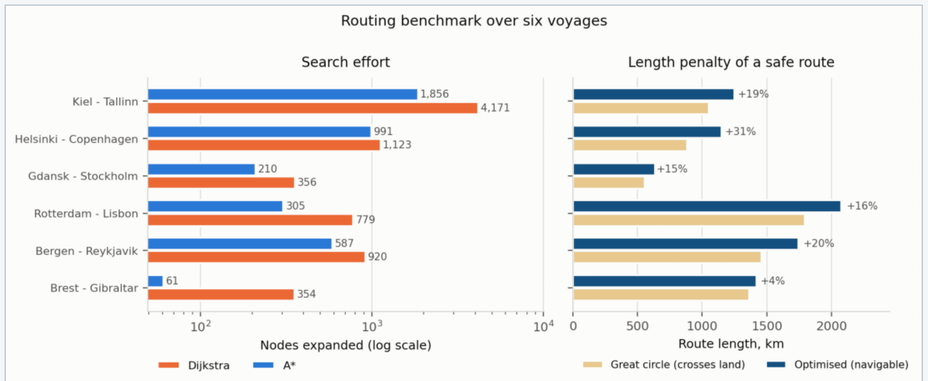
$$c(u,v) \;=\; d_{\mathrm{gc}}(u,v)\cdot\frac{w(z_U)+w(z_V)}{2}$$

where d_{gc} is the great-circle length of the step and $w(z)$ the weight of the zone the cell was classified into. The A* heuristic

$$h(v) \;=\; w_{\min}\cdot d_{\mathrm{gc}}(v,\ \mathrm{goal})$$

never over-estimates the remaining cost, because no cell can be cheaper than w_{min}. The heuristic is therefore admissible and A* returns the same optimum as Dijkstra — confirmed experimentally on every test voyage (maximum cost deviation 0.0 %).

Comparison with the shortest route



Complexity and measured performance

Property	Dijkstra	A*
Time complexity	O(E log V)	O(E log V), pruned by h
Space complexity	O(V)	O(V)
Optimal result	yes	yes (h admissible)
Mean nodes expanded	1,283	668
Relative search effort	1.00×	0.40×
Relative search time	1.00×	0.81×

Notes on the implementation

- Diagonal steps are allowed (8-connected lattice); each step length is the true great-circle distance for that row, pre-computed once per offset so the inner loop contains no trigonometry.

- The A* heuristic is evaluated for every generated node and is therefore written with pre-computed per-row trigonometry; replacing a NumPy call with plain `math` made A* 1.8× faster and reversed the measured ranking.

Application for constructing a maritime vessel's course based on neural-network classification of geographic zones

Diploma project · Student: Li Haoran

Brest State Technical University · Faculty EIS, Department IIT · 2026/27